Lab 7: Fuzzing for Input Validation Bugs (Hypothesis)

Name: Priyanka B M SRN: PES1UG24AM207 Repo: SETAPESU26/L6_L7_25_train_fare (Lab 7)

Step 1: Installing dependencies

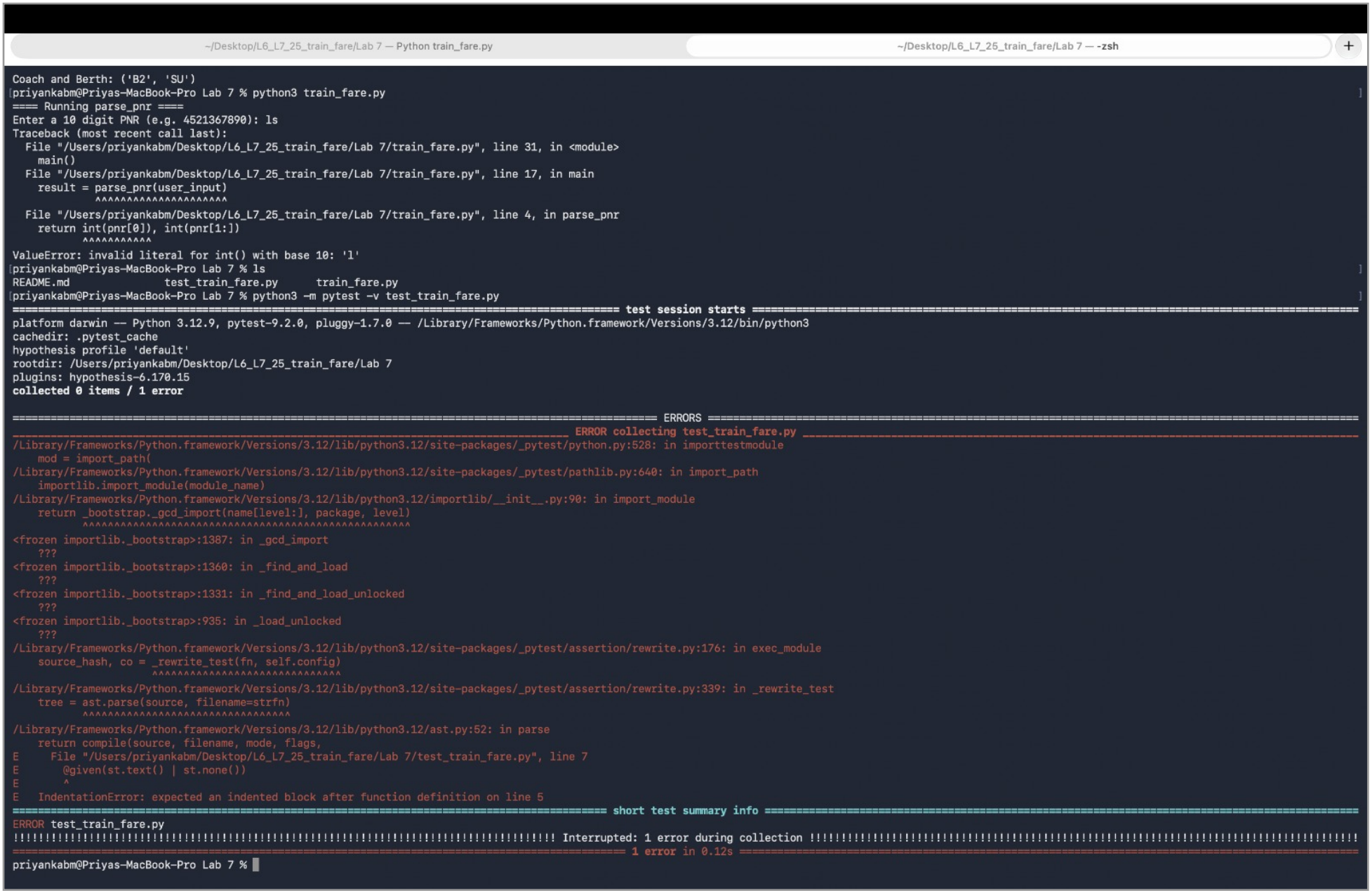
pip3 install hypothesis pytest (already satisfied), then python3 train_fare.py started

```
priyankabm@Priyas-MacBook-Pro Lab 7 % pip3 install hypothesis pytest
[python3 train_fare.py
Requirement already satisfied: hypothesis in /Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages (6.170.15)
Requirement already satisfied: pytest in /Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages (9.2.0)
Requirement already satisfied: sortedcontainers<3.0.0,>=2.1.0 in /Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages (from hypothesis) (2.4.0)
Requirement already satisfied: iniconfig>=2 in /Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages (from pytest) (2.3.1)
Requirement already satisfied: packaging>=24 in /Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages (from pytest) (26.3)
Requirement already satisfied: pluggy<2,>=1.6 in /Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages (from pytest) (1.7.0)
Requirement already satisfied: pygments>=2.15 in /Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages (from pytest) (2.21.0)

[notice] A new release of pip is available: 24.3.1 -> 26.3
[notice] To update, run: pip3 install --upgrade pip
==== Running parse_pnr ====
Enter a 10 digit PNR (e.g. 4521367890):
```

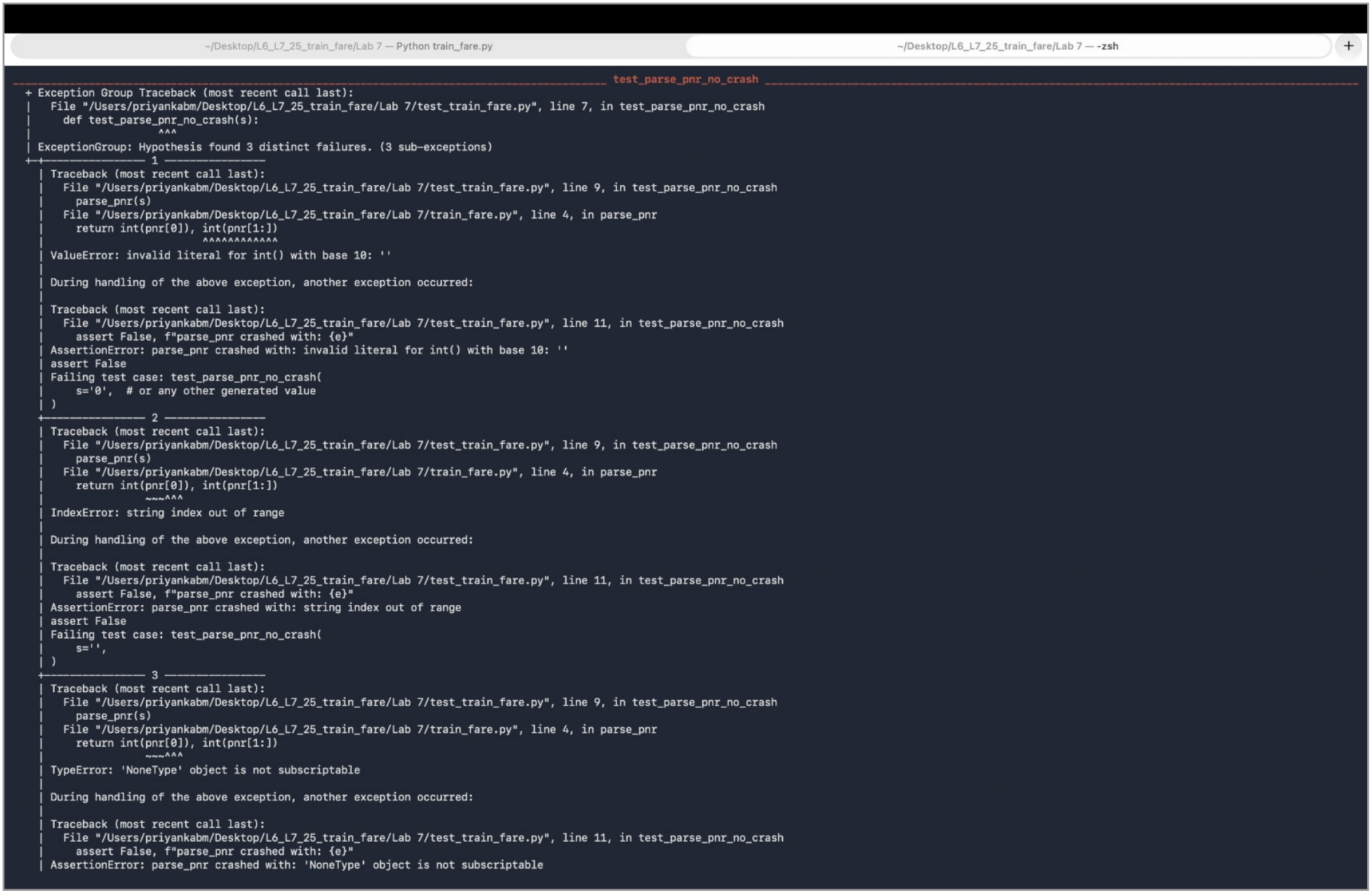
Step 2: Running the buggy application

python3 train_fare.py crashes with a ValueError on invalid input (original code, before the fix)



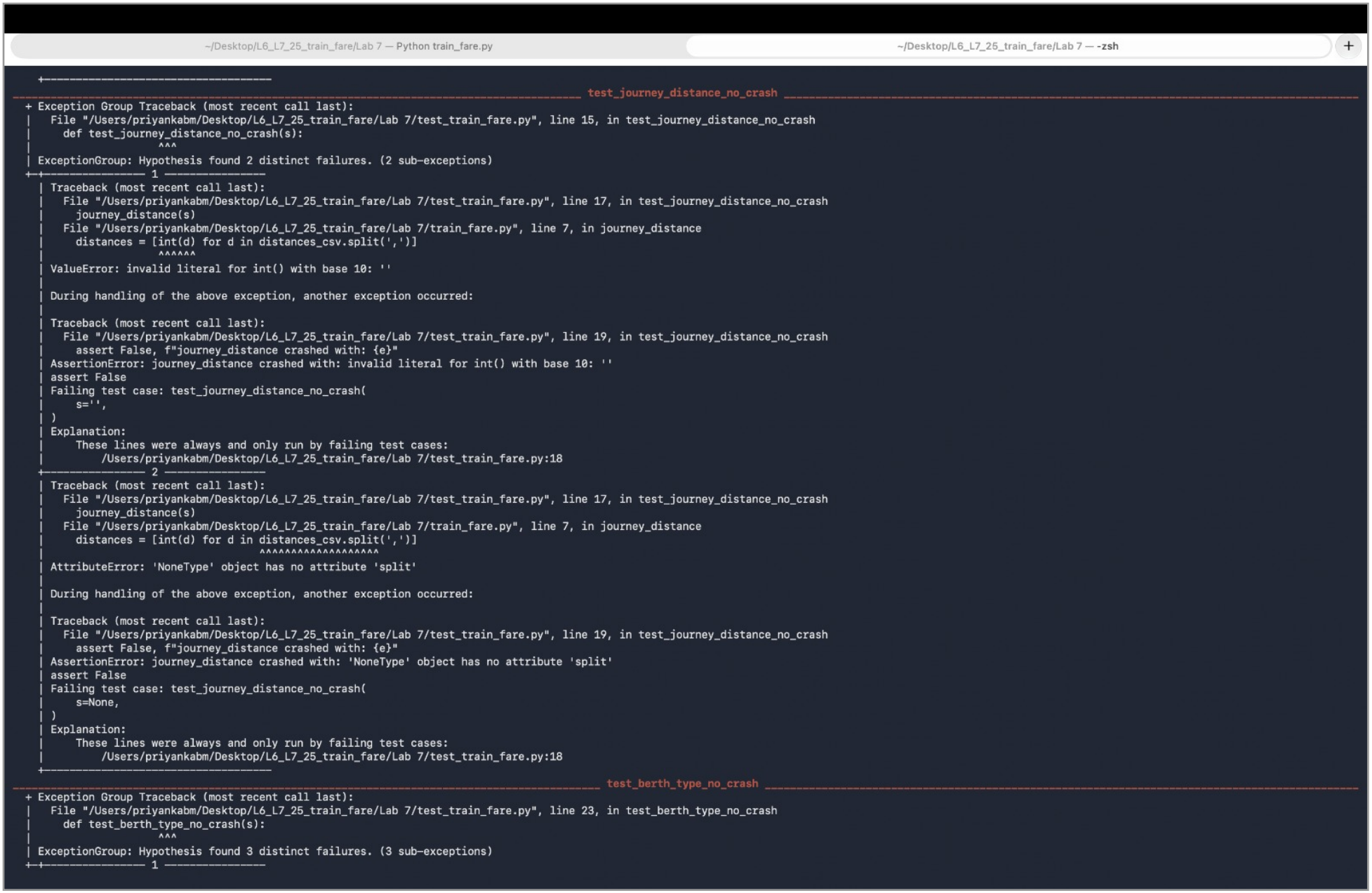
Step 4: Running test_train_fare.py on the buggy code (1 of 4)

test_parse_pnr_no_crash: Hypothesis finds 3 distinct failures ('0', '', None)



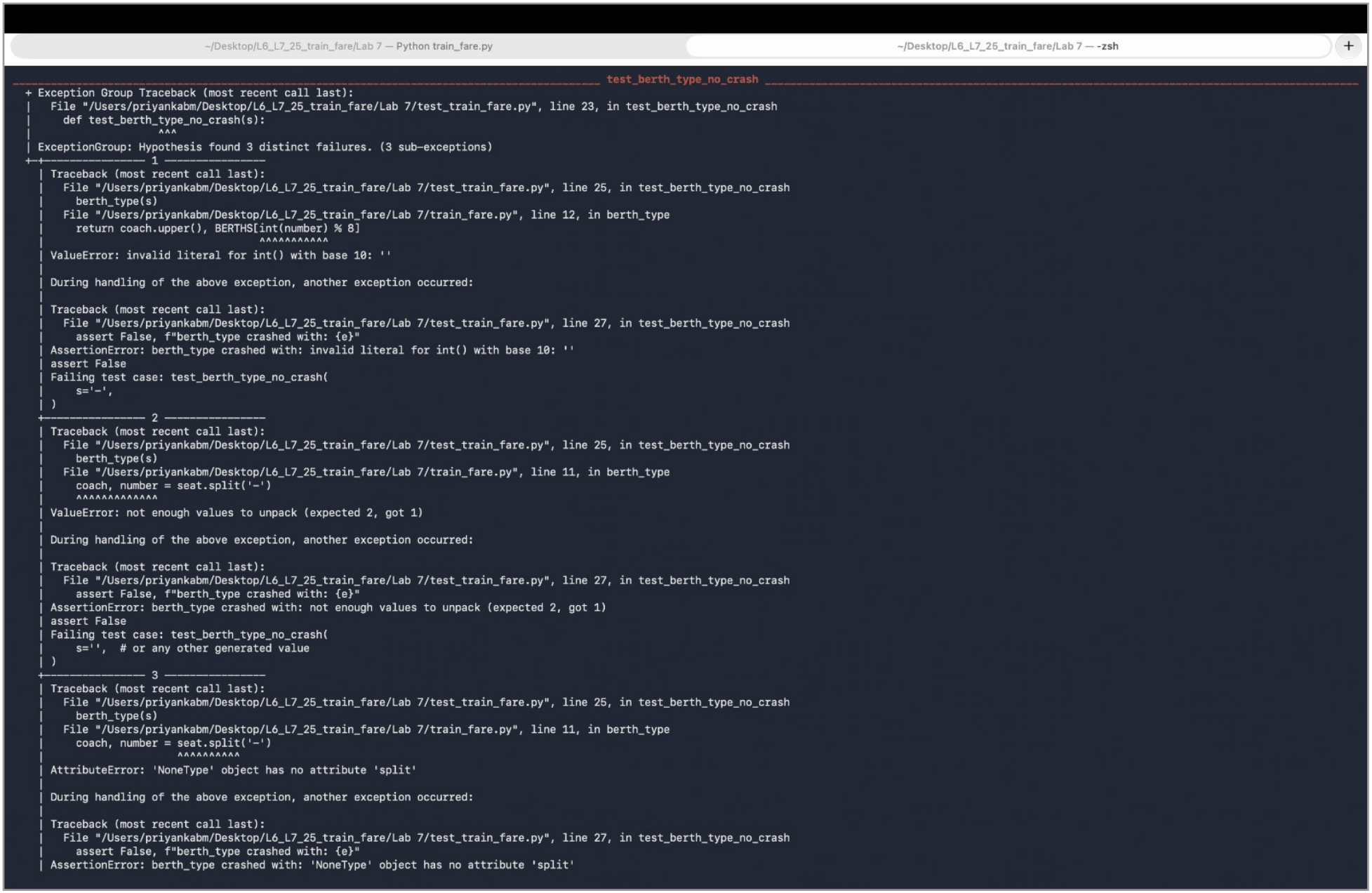
Step 4: Running test_train_fare.py on the buggy code (2 of 4)

test_journey_distance_no_crash: 2 distinct failures (" and None)



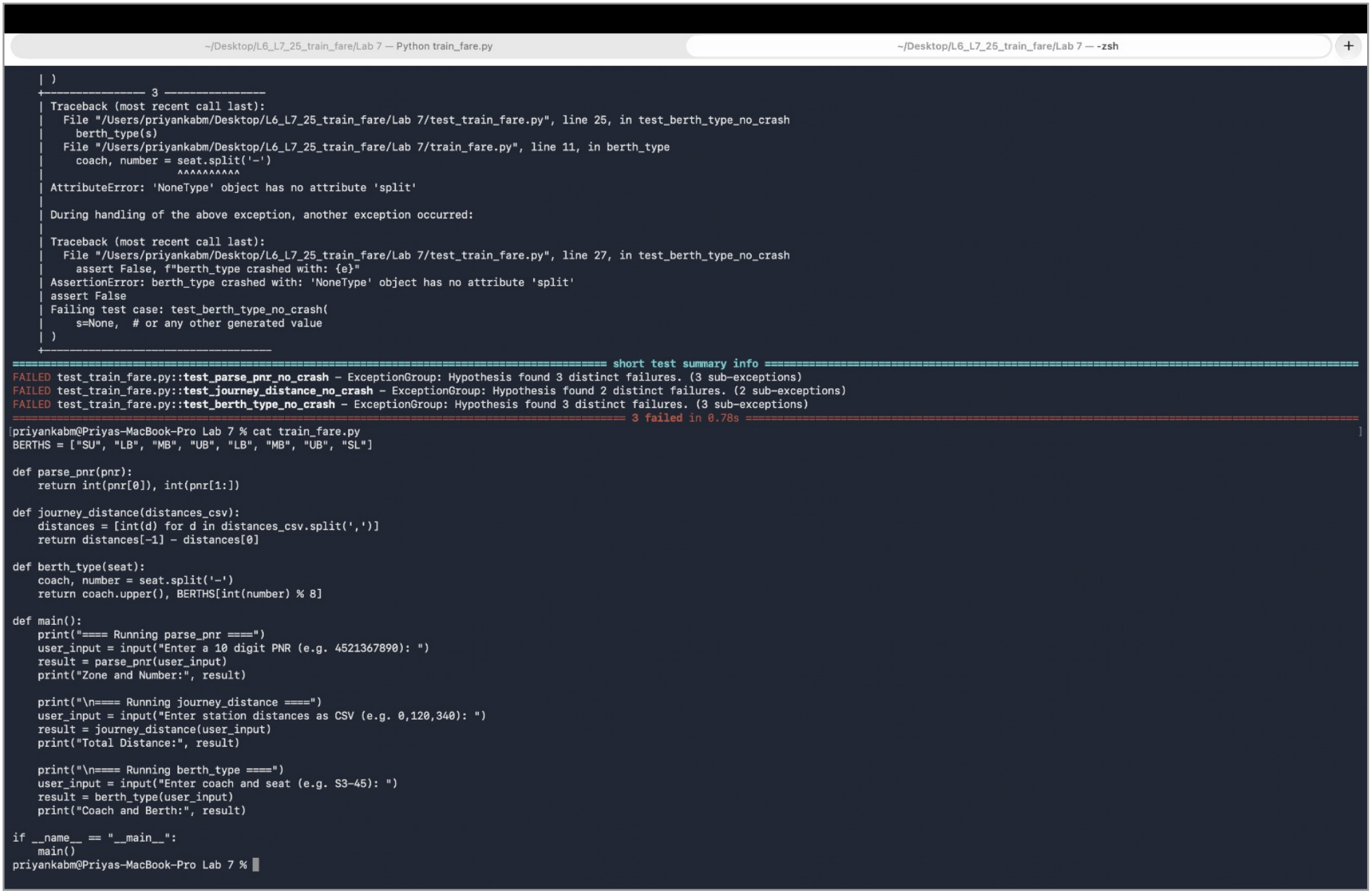
Step 4: Running test_train_fare.py on the buggy code (3 of 4)

test_berth_type_no_crash: 3 distinct failures ('-', '' and None)



Step 4: Running test_train_fare.py on the buggy code (4 of 4)

Summary: 3 failed. The original train_fare.py is shown below the summary.



Step 5: Running the fixed application

python3 train_fare.py after hardening the functions

```
    main()
priyankabm@Priyas-MacBook-Pro Lab 7 % python3 train_fare.py
==== Running parse_pnr ====
Enter a 10 digit PNR (e.g. 4521367890): 9876543210
Zone and Number: (9, 876543210)

==== Running journey_distance ====
Enter station distances as CSV (e.g. 0,120,340): 100,250,600,900
Total Distance: 800

==== Running berth_type ====
Enter coach and seat (e.g. S3-45): S5-8
Coach and Berth: ('S5', 'SU')
priyankabm@Priyas-MacBook-Pro Lab 7 %
```

Step 6: Re-running test_train_fare.py on the fixed code

python3 -m pytest -v test_train_fare.py: all 3 tests passed

